



Strings:

1. Reverse a String

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    string str = "hello";
    reverse(str.begin(), str.end());
    cout << "Reversed: " << str;
    return 0;
}
```

2. Palindrome Check

```
#include <bits/stdc++.h>
using namespace std;

bool isPalindrome(string str) {
    int left = 0, right = str.size() - 1;
    while (left < right) {
        if (str[left] != str[right]) return false;
    }
}
```

```

        left++; right--;
    }
    return true;
}

int main() {
    string str = "madam";
    cout << (isPalindrome(str) ? "Palindrome" : "Not Palindrome");
    return 0;
}

```

3. Count Vowels and Consonants

```

#include <bits/stdc++.h>
using namespace std;

int main() {
    string str = "hello world";
    int vow = 0, cons = 0;

    for (char ch : str) {
        ch = tolower(ch);
        if (isalpha(ch)) {
            if (ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u') vow++;
            else cons++;
        }
    }
    cout << "Vowels: " << vow << ", Consonants: " << cons;
    return 0;
}

```

4. Character Frequency

```

#include <bits/stdc++.h>
using namespace std;

int main() {

```

```

string str = "programming";
map<char,int> freq;

for (char ch : str) freq[ch]++;

for (auto p : freq)
    cout << p.first << " : " << p.second << endl;
return 0;
}

```

5. Common Prefix

```

#include <bits/stdc++.h>
using namespace std;

string commonPrefix(vector<string>& arr) {
    if (arr.empty()) return "";
    string pre = arr[0];
    for (int i = 1; i < arr.size(); i++) {
        while (arr[i].find(pre) != 0)
            pre = pre.substr(0, pre.size() - 1);
    }
    return pre;
}

int main() {
    vector<string> arr = {"flower","flow","flight"};
    cout << "Common Prefix: " << commonPrefix(arr);
    return 0;
}

```

6. Anagram Check

```

#include <bits/stdc++.h>
using namespace std;

bool isAnagram(string s1, string s2) {

```

```

    if (s1.size() != s2.size()) return false;
    sort(s1.begin(), s1.end());
    sort(s2.begin(), s2.end());
    return s1 == s2;
}

int main() {
    string s1 = "listen", s2 = "silent";
    cout << (isAnagram(s1, s2) ? "Anagram" : "Not Anagram");
    return 0;
}

```

Array Questions

1. Find Maximum and Minimum Element in Array

```

#include <iostream>
using namespace std;

int main() {
    int arr[] = {10, 20, 4, 45, 99};
    int size = 5;

    int maxVal = arr[0], minVal = arr[0];
    for (int i = 1; i < size; i++) {
        if (arr[i] > maxVal) maxVal = arr[i];
        if (arr[i] < minVal) minVal = arr[i];
    }

    cout << "Maximum: " << maxVal << ", Minimum: " << minVal;
    return 0;
}

```

3. Reverse an Array

```

#include <iostream>
using namespace std;

int main() {
    int arr[] = {1, 2, 3, 4, 5};
    int size = 5;

    int start = 0, end = size - 1;
    while (start < end) {
        int temp = arr[start];
        arr[start] = arr[end];
        arr[end] = temp;
        start++;
        end--;
    }

    cout << "Reversed Array: ";
    for (int i = 0; i < size; i++) cout << arr[i] << " ";
    return 0;
}

```

4. Check if Array is Sorted

```

#include <iostream>
using namespace std;

int main() {
    int arr[] = {1, 2, 3, 5, 6};
    int size = 5;

    bool sorted = true;
    for (int i = 1; i < size; i++) {
        if (arr[i] < arr[i - 1]) {
            sorted = false;
            break;
        }
    }
}

```

```
if (sorted) cout << "Array is Sorted";  
else cout << "Array is Not Sorted";  
return 0;  
}
```

5. Find Missing Number (1 to n)

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int arr[] = {1, 2, 4, 5, 6};  
    int size = 5;  
  
    int n = size + 1;  
    int totalSum = n * (n + 1) / 2;  
    int currentSum = 0;  
  
    for (int i = 0; i < size; i++) currentSum += arr[i];  
  
    cout << "Missing Number: " << totalSum - currentSum;  
    return 0;  
}
```

6. Find Duplicate Elements

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int arr[] = {1, 3, 2, 3, 4, 5, 2};  
    int size = 7;  
  
    cout << "Duplicate elements: ";  
    bool found = false;
```

```

for (int i = 0; i < size; i++) {
    for (int j = i + 1; j < size; j++) {
        if (arr[i] == arr[j]) {
            cout << arr[i] << " ";
            found = true;
            break;
        }
    }
}

if (!found) cout << "None";
return 0;
}

```

7. Count Frequency of Elements

```

#include <iostream>
using namespace std;

int main() {
    int arr[] = {1, 2, 2, 3, 4, 1, 2};
    int size = 7;
    bool visited[7] = {false};

    for (int i = 0; i < size; i++) {
        if (visited[i]) continue;
        int count = 1;
        for (int j = i + 1; j < size; j++) {
            if (arr[i] == arr[j]) {
                visited[j] = true;
                count++;
            }
        }
        cout << arr[i] << " → " << count << endl;
    }
    return 0;
}

```

8. Rotate Array Right by 1

```
#include <iostream>
using namespace std;

int main() {
    int arr[] = {1, 2, 3, 4, 5};
    int size = 5;

    int last = arr[size - 1];
    for (int i = size - 1; i > 0; i--) {
        arr[i] = arr[i - 1];
    }
    arr[0] = last;

    cout << "Rotated Array: ";
    for (int i = 0; i < size; i++) cout << arr[i] << " ";
    return 0;
}
```

10. Find Pairs with Given Sum

```
#include <iostream>
using namespace std;

int main() {
    int arr[] = {1, 4, 5, 3, 2};
    int size = 5;
    int target = 6;

    cout << "Pairs with sum " << target << ": ";
    bool found = false;

    for (int i = 0; i < size; i++) {
        for (int j = i + 1; j < size; j++) {
            if (arr[i] + arr[j] == target) {
                cout << "(" << arr[i] << ", " << arr[j] << ") ";
            }
        }
    }
}
```



```

        found = true;
    }
}
}

if (!found) cout << "None";
return 0;
}

```

11. Merge Two Sorted Arrays

```

#include <iostream>
using namespace std;

int main() {
    int arr1[] = {1, 3, 5};
    int arr2[] = {2, 4, 6};
    int size1 = 3, size2 = 3;

    int merged[size1 + size2];
    int i = 0, j = 0, k = 0;

    while (i < size1 && j < size2) {
        if (arr1[i] < arr2[j]) merged[k++] = arr1[i++];
        else merged[k++] = arr2[j++];
    }
    while (i < size1) merged[k++] = arr1[i++];
    while (j < size2) merged[k++] = arr2[j++];

    cout << "Merged Array: ";
    for (int m = 0; m < size1 + size2; m++) cout << merged[m] << " ";
    return 0;
}

```

15. Majority Element (> n/2 times)

```

#include <iostream>
using namespace std;

int main() {
    int arr[] = {2, 2, 1, 2, 3, 2, 2};
    int size = 7;

    int majority = -1;
    for (int i = 0; i < size; i++) {
        int count = 1;
        for (int j = i + 1; j < size; j++) {
            if (arr[i] == arr[j]) count++;
        }
        if (count > size / 2) {
            majority = arr[i];
            break;
        }
    }

    if (majority != -1) cout << "Majority Element: " << majority;
    else cout << "No Majority Element";
    return 0;
}

```

Math Questions

1. Check Prime Number

```

#include <iostream>
using namespace std;

int main() {
    int num = 29;
    bool prime = true;

    if (num < 2) prime = false;
}

```

```

else {
    for (int i = 2; i * i <= num; i++) {
        if (num % i == 0) {
            prime = false;
            break;
        }
    }
}

if (prime) cout << num << " is Prime";
else cout << num << " is Not Prime";
return 0;
}

```

2. Find GCD and LCM

```

#include <iostream>
using namespace std;

int main() {
    int a = 12, b = 18;
    int x = a, y = b;

    while (b != 0) {
        int temp = b;
        b = a % b;
        a = temp;
    }

    int gcd = a;
    int lcm = (x * y) / gcd;

    cout << "GCD: " << gcd << ", LCM: " << lcm;
    return 0;
}

```

3. Check Armstrong Number

```

#include <iostream>
using namespace std;

int main() {
    int num = 153, temp = num, sum = 0, digits = 0;

    // count digits
    while (temp > 0) {
        digits++;
        temp /= 10;
    }

    temp = num;
    while (temp > 0) {
        int digit = temp % 10;
        int power = 1;
        for (int i = 0; i < digits; i++) {
            power *= digit;
        }
        sum += power;
        temp /= 10;
    }

    if (sum == num) cout << num << " is Armstrong";
    else cout << num << " is Not Armstrong";
    return 0;
}

```

4. Check Palindrome Number

```

#include <iostream>
using namespace std;

int main() {
    int num = 1221, original = num, reversed = 0;

    while (num > 0) {

```

```

        reversed = reversed * 10 + (num % 10);
        num /= 10;
    }

    if (original == reversed) cout << original << " is Palindrome";
    else cout << original << " is Not Palindrome";
    return 0;
}

```

5. Factorial of a Number

```

#include <iostream>
using namespace std;

int main() {
    int num = 5;
    int fact = 1;

    for (int i = 2; i <= num; i++) {
        fact *= i;
    }

    cout << "Factorial of " << num << " is " << fact;
    return 0;
}

```

6. Fibonacci Series

```

#include <iostream>
using namespace std;

int main() {
    int n = 10;
    int a = 0, b = 1;

    cout << "Fibonacci Series: ";
    for (int i = 0; i < n; i++) {

```

```
    cout << a << " ";  
    int next = a + b;  
    a = b;  
    b = next;  
}  
return 0;  
}
```

7. Perfect Number

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int num = 28, sum = 0;  
  
    for (int i = 1; i <= num / 2; i++) {  
        if (num % i == 0) sum += i;  
    }  
  
    if (sum == num) cout << num << " is Perfect Number";  
    else cout << num << " is Not Perfect Number";  
    return 0;  
}
```

9. Count Digits in a Number

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int num = 12345, count = 0;  
  
    while (num > 0) {  
        count++;  
        num /= 10;  
    }  
}
```

```
    cout << "Number of digits: " << count;
    return 0;
}
```

10. Reverse a Number

```
#include <iostream>
using namespace std;

int main() {
    int num = 12345, rev = 0;

    while (num > 0) {
        rev = rev * 10 + (num % 10);
        num /= 10;
    }

    cout << "Reversed Number: " << rev;
    return 0;
}
```

11. Sum of Digits

```
#include <iostream>
using namespace std;

int main() {
    int num = 1234, sum = 0;

    while (num > 0) {
        sum += (num % 10);
        num /= 10;
    }

    cout << "Sum of Digits: " << sum;
```

```
    return 0;
}
```

12. Power of a Number (x^y)

```
#include <iostream>
using namespace std;

int main() {
    int base = 2, exp = 5;
    int result = 1;

    for (int i = 0; i < exp; i++) {
        result *= base;
    }

    cout << base << "^" << exp << " = " << result;
    return 0;
}
```

13. Find the Largest Element in an Array

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter size of array: ";
    cin >> n;

    int arr[n];
    cout << "Enter " << n << " elements: ";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }

    int largest = arr[0]; // assume first element is largest
```



```
for (int i = 1; i < n; i++) {  
    if (arr[i] > largest) {  
        largest = arr[i];  
    }  
}  
  
cout << "Largest element: " << largest << endl;  
return 0;  
}
```